www.XtremePapers.com

UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

**BIOLOGY** **5090/13**

Paper 1 Multiple Choice **May/June 2010**

**1 hour**

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

*0458193692*

**READ THESE INSTRUCTIONS FIRST**

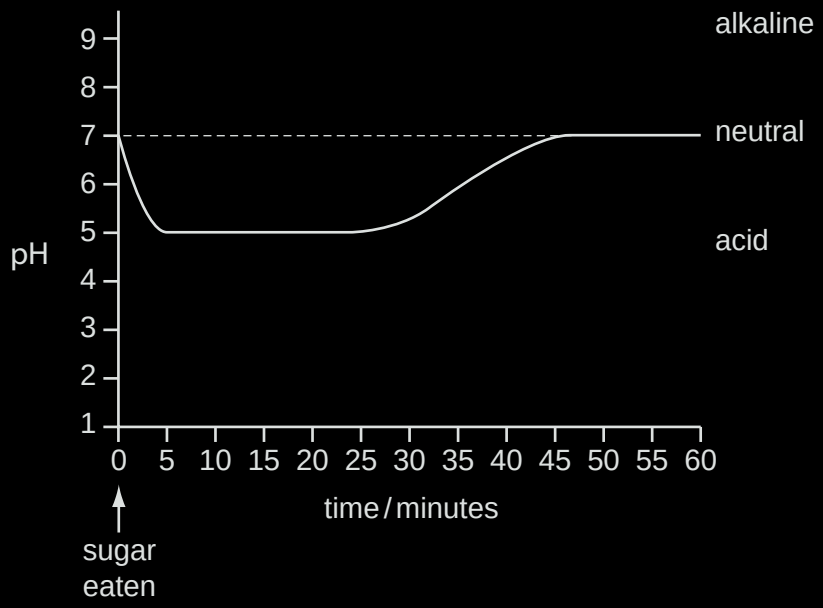
Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

**Read the instructions on the Answer Sheet very carefully.**

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

UNIVERSITY of CAMBRIDGE
International Examinations




When are conditions in the mouth most likely to cause tooth decay?

**A** 0-5 minutes

**B** 5-25 minutes

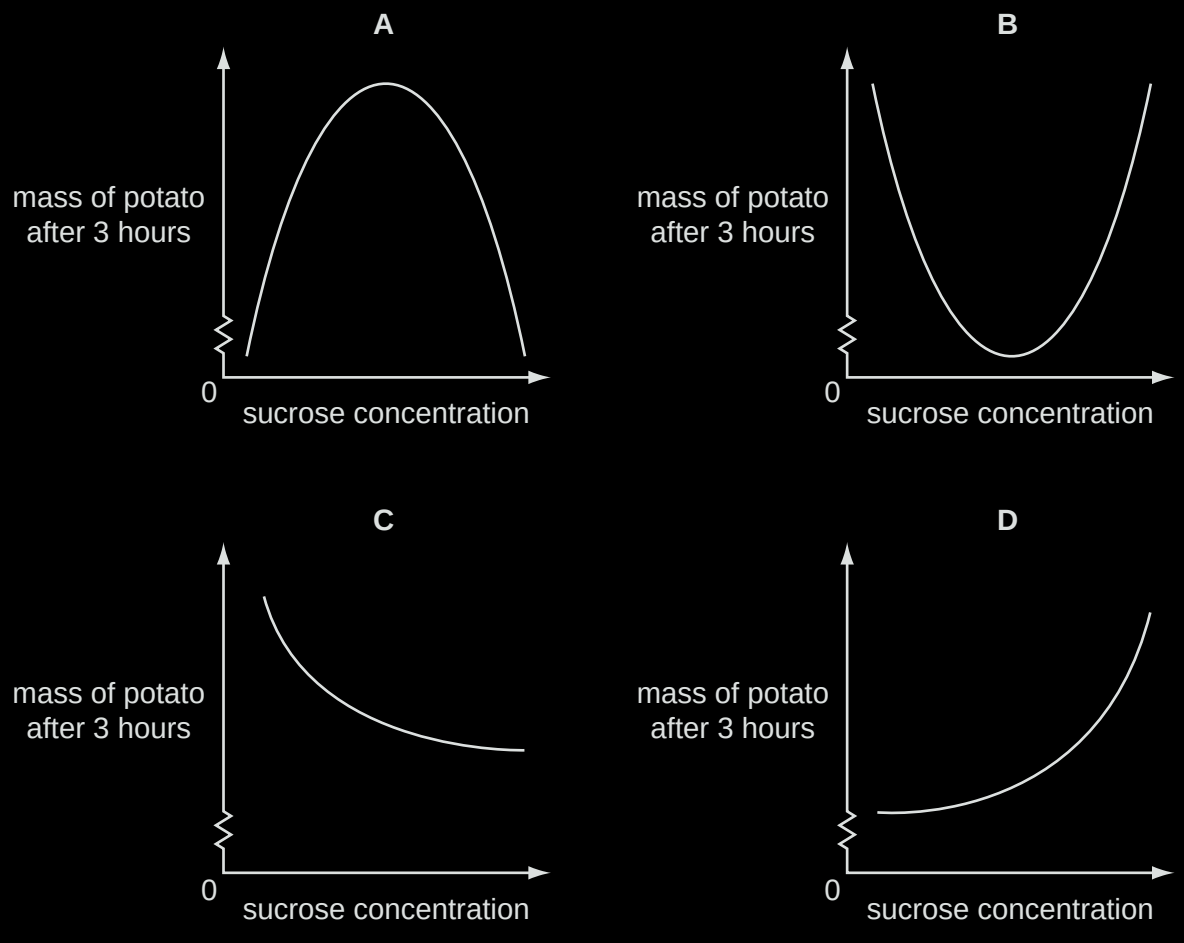
**C** 25-45 minutes

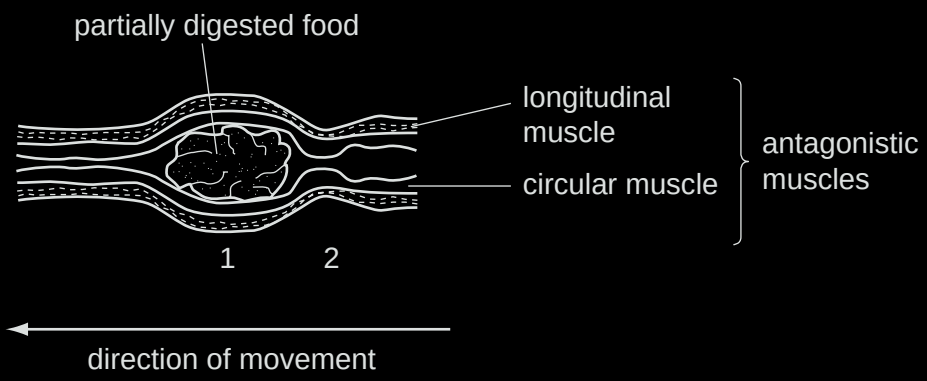
**D** 45-60 minutes

**2** The table shows the recommended daily intake of energy and of iron for females at four different ages: 6 months, 5 years, 25 years and 75 years.

Which shows the recommended daily intake for the 25 year old?

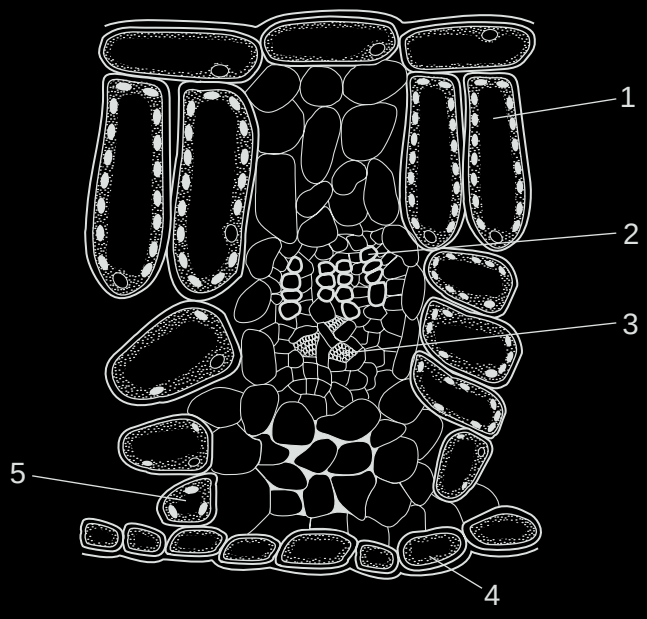
| | energy / MJ | iron / mg |
|---|---|---|
| **A** | 2.7 | 4.3 |
| **B** | 6.5 | 6.1 |
| **C** | 7.6 | 8.7 |
| **D** | 8.1 | 14.8 |


A
mass of potato after 3 hours
0
sucrose concentration
B
mass of potato after 3 hours
0
sucrose concentration
C
mass of potato after 3 hours
0
sucrose concentration
D
mass of potato after 3 hours
0
sucrose concentration




What is the state of the longitudinal muscles at 1and 2?

| | 1 | 2 |
|---|---|---|
| **A** | contracted | contracted |
| **B** | contracted | relaxed |
| **C** | relaxed | contracted |
| **D** | relaxed | relaxed |




Which cells have the functions shown?

| | photosynthesis | transport |
|---|---|---|
| **A** | 1 and 5 | 2 and 3 |
| **B** | 2 and 4 | 1 and 4 |
| **C** | 3 and 4 | 2 and 5 |
| **D** | 4 and 5 | 3 and 4 |

**9** Which substances are transported in xylem and in phloem?

| | xylem | phloem |
|---|---|---|
| **A** | amino acids and mineral ions | amino acids and water |
| **B** | mineral ions and sucrose | starch and mineral ions |
| **C** | mineral ions and water | sucrose and water |
| **D** | starch and water | sucrose and starch |

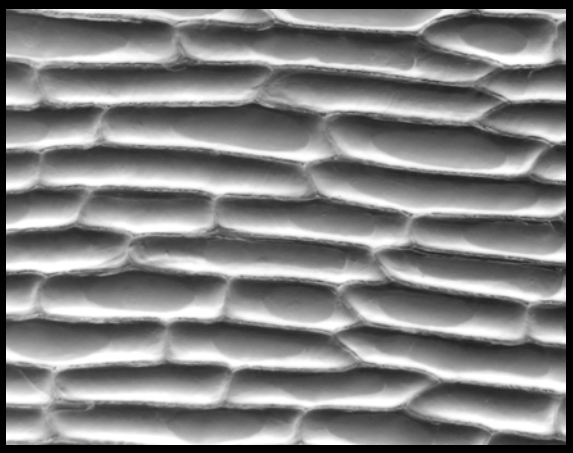
**10** Which signs show that a plant has been grown in a culture solution deficient in magnesium?

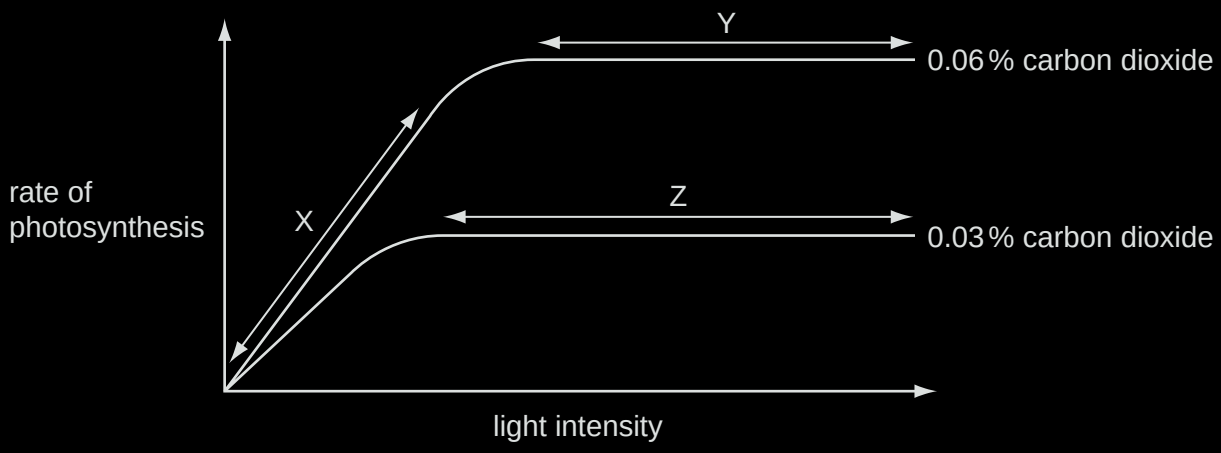
**A** purple leaves and poor root growth

**B** small leaves and thin stem

**C** white upper leaves and normal lower leaves

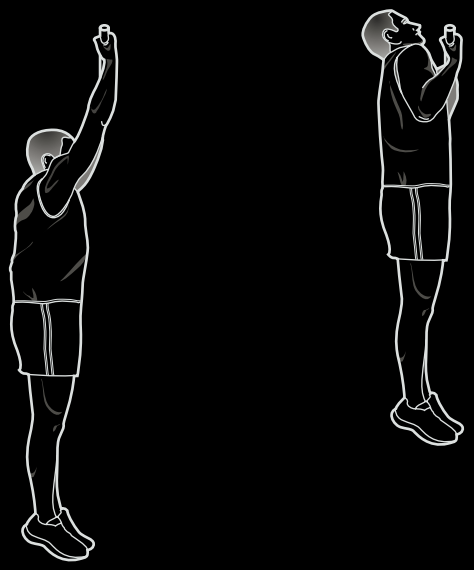
**D** yellow stem and yellow leaves






What may be limiting the rate of photosynthesis at X, Y and Z?

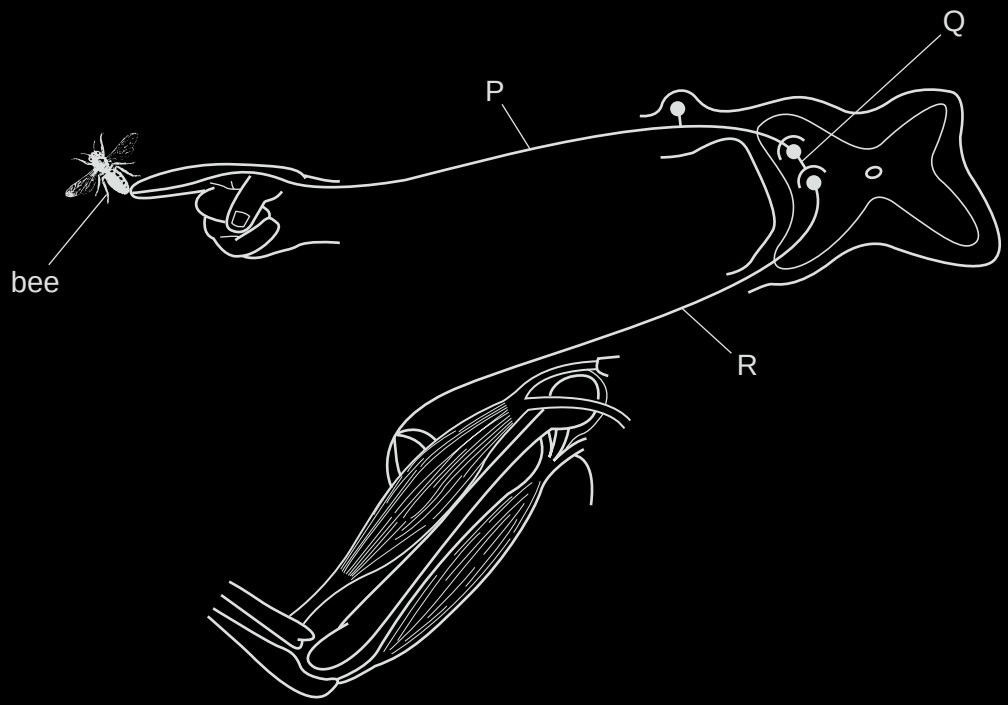
| | X | Y | Z |
|---|---|---|---|
| **A** | carbon dioxide | light intensity | carbon dioxide |
| **B** | carbon dioxide | light intensity | light intensity |
| **C** | light intensity | carbon dioxide | carbon dioxide |
| **D** | light intensity | carbon dioxide | light intensity |



X Y

In moving from the lower position X to the upper position Y, which muscles and joint movements occur?

| | biceps | triceps | ball and socket joint |
|---|---|---|---|
| **A** | contracts | relaxes | rotates |
| **B** | contracts | relaxes | extends |
| **C** | relaxes | contracts | rotates |
| **D** | relaxes | contracts | flexes |



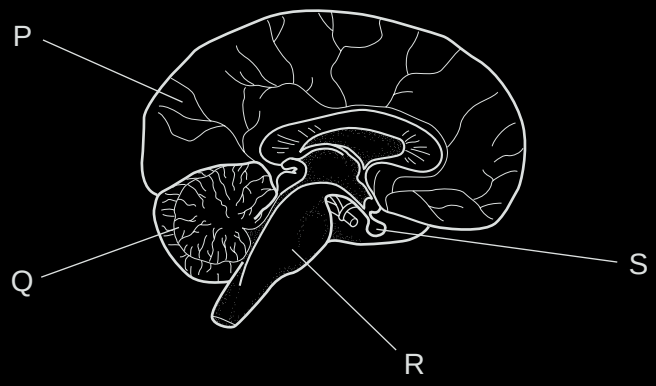

If the relay neurone at Q is damaged, how will the transmission of nerve impulses in the reflex arc be affected?

**A** They cannot pass from P to Q.

**B** They cannot pass from P to R.

**C** They cannot pass from Q to P.

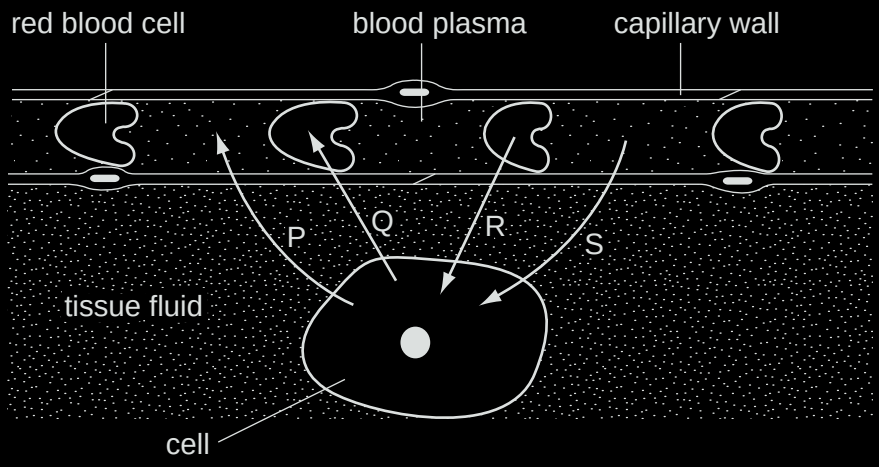
**D** They cannot pass from R to Q.




Which row correctly identifies the cerebral hemisphere, cerebellum, medulla and pituitary gland?

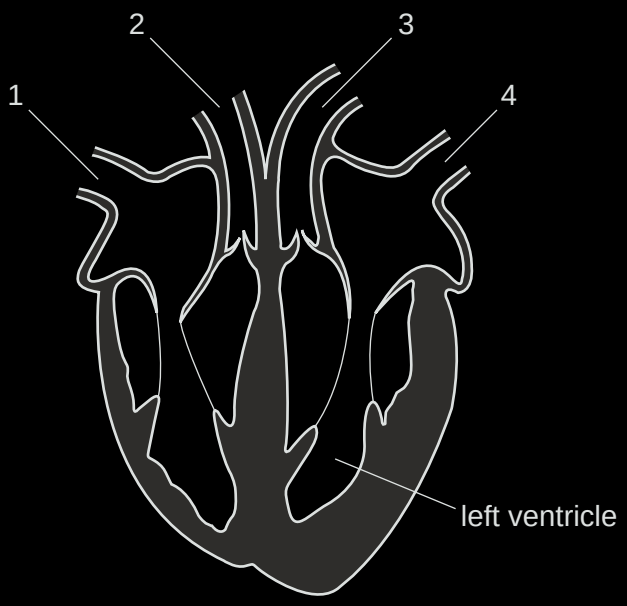
| | cerebral hemisphere | cerebellum | medulla | pituitary gland |
|---|---|---|---|---|
| **A** | P | Q | R | S |
| **B** | P | R | Q | S |
| **C** | Q | R | S | P |
| **D** | S | P | R | Q |

**16** The diagram represents a blood capillary with an adjacent cell. The arrows represent the transfer of substances between the capillary and the cell.




Which arrows represent glucose, carbon dioxide and oxygen?

| | glucose | carbon dioxide | oxygen |
|---|---|---|---|
| **A** | P | R | Q |
| **B** | Q | S | P |
| **C** | R | Q | S |
| **D** | S | P | R |




What are the functions of the numbered blood vessels?

| | carries blood to body | carries blood to lungs | carries blood from lungs | carries blood from body |
|---|---|---|---|---|
| **A** | 1 | 4 | 3 | 2 |
| **B** | 2 | 1 | 3 | 4 |
| **C** | 2 | 3 | 4 | 1 |
| **D** | 3 | 2 | 4 | 1 |

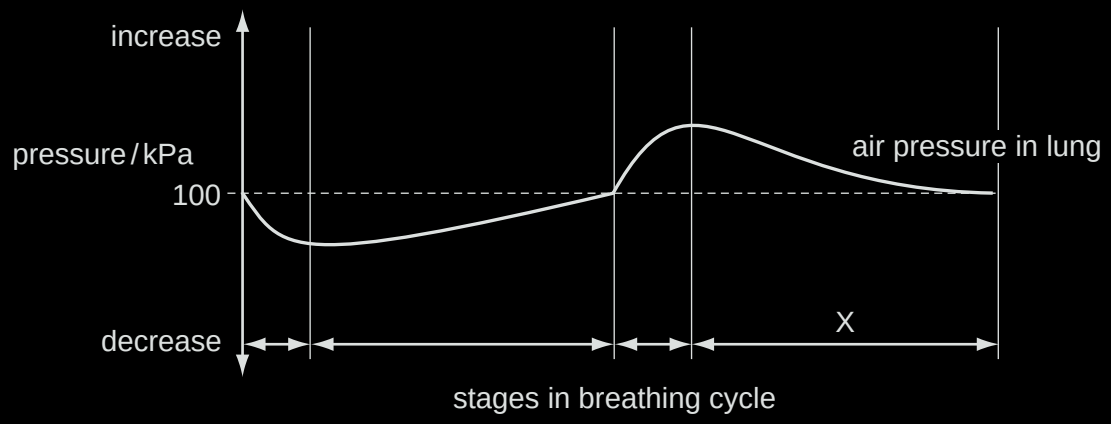
**18** Which structures are **all** involved in controlling human body temperature?

**A** blood vessels near the skin surface, the cerebellum and sweat glands

**B** blood vessels near the skin surface, the hypothalamus and skeletal muscles

**C** kidneys, the cerebellum and sweat glands

**D** kidneys, the hypothalamus and skeletal muscles




What causes the change in air pressure during stage X?

**A** contraction of diaphragm muscles

**B** increase in volume of lungs

**C** outflow of air from lungs

**D** relaxation of internal intercostal muscles

**20** What do the cilia do in the bronchi of the lungs?

| | trap bacteria | move mucus out of the lungs |
|---|---|---|
| **A** | ✓ | ✓ |
| **B** | ✓ | ✗ |
| **C** | ✗ | ✓ |
| **D** | ✗ | ✗ |

key

✓ = function of the cilia

✗ = not a function of the cilia

**21** Fitness training increases the concentration of lactic acid that runners can build up in their muscles before pain stops them running.

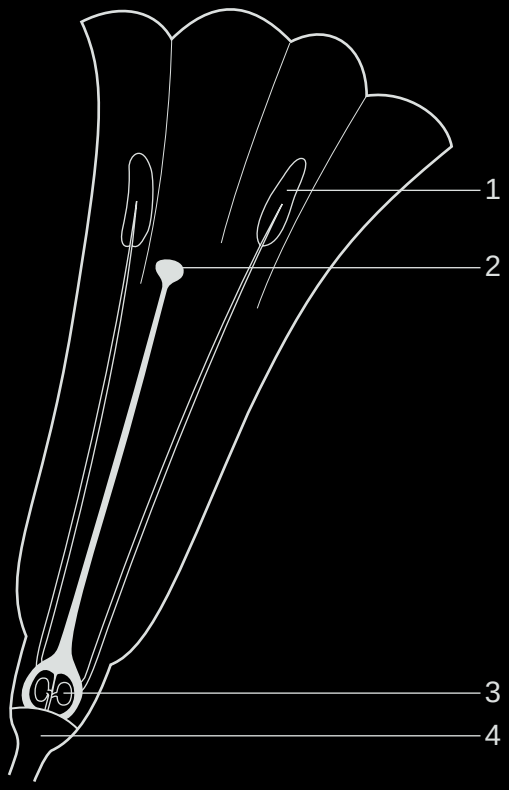
What is a consequence of this increase?

**A** Aerobic respiration in muscles can be more rapid.

**B** Blood flow to the muscles is increased.

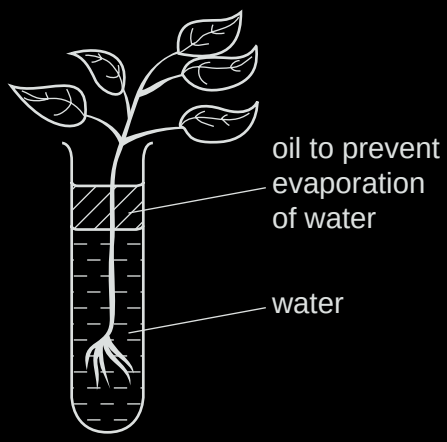
**C** More anaerobic respiration can take place in muscles.

**D** More energy is needed by the muscles.




In which structures are haploid nuclei formed by reduction division?

**A** 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4




Some of the plants have their leaves coated with grease to reduce transpiration. Each plant is weighed in its test-tube at the start of the experiment and again two days later.

The results are shown in the table.

| | mass / g | | | | |
|---|---|---|---|---|---|
| | plant 1 | plant 2 | plant 3 | plant 4 | plant 5 |
| at the start of experiment | 105 | 121 | 107 | 111 | 119 |
| after two days | 103 | 97 | 84 | 110 | 93 |

Which plants had their leaves coated with grease?

**A** 1 and 2 **B** 1 and 4 **C** 2 and 5 **D** 2, 3 and 5

**26** Which statement about human blood groups is correct?

**A** A person with the blood group A cannot have an $I^O$ allele.

**B** A person with the blood group B may have either the genotype $I^BI^B$ or $I^BI^O$.

**C** In a person with blood group AB, the $I^B$ allele is recessive to the allele $I^A$.

**D** The alleles $I^O$ and $I^B$ are co-dominant and have equal effect on a phenotype.

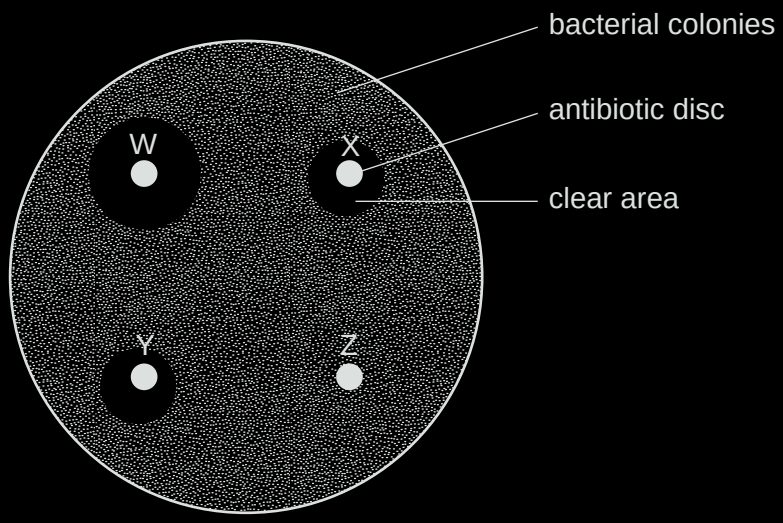
**27** Which statement about chromosomes is correct?

**A** Chromosomes are long DNA molecules called genes which are divided into sections.

**B** Chromosomes include a long molecule of DNA divided into sections called genes.

**C** Chromosomes include genes which are divided into sections called DNA molecules.

**D** Genes include long DNA molecules called chromosomes.




Which conclusion about this experiment is correct?

**A** Antibiotic W is more effective against these bacteria than antibiotic X.

**B** Antibiotic Y is more effective against these bacteria than antibiotic W.

**C** Disc W has a higher concentration of antibiotic than disc Y.

**D** Disc Z contains no antibiotic.



In what way does this differ from a cell of a fungus?

**A** The bacterium has a cell membrane.

**B** The bacterium has a cell wall.

**C** The bacterium has cytoplasm.

**D** The bacterium has no true nucleus.

**31** Bacteria can be genetically modified to produce human insulin.

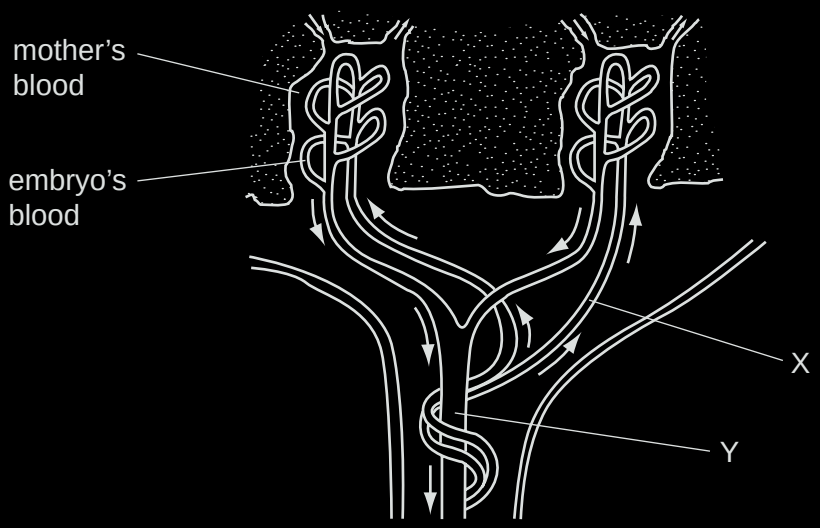
What is a possible danger of this procedure?

**A** Bacterial insulin is less effective in treating diabetes than animal insulin.

**B** The genetically modified bacteria may become insulin resistant.

**C** The genetically modified bacteria may produce too much insulin.

**D** The presence of a new gene in the bacteria may alter the way that existing genes work.




Which substances are present at X in higher concentrations than at Y?

**A** carbon dioxide and glucose

**B** carbon dioxide and urea

**C** glucose and oxygen

**D** glucose and urea

**33** After sexual intercourse, sperm can survive for 3 days in the uterus and oviducts. Ovulation can occur any time from day 13 to day 15 and an egg cell can live for 2 days after ovulation.

How long is the longest possible fertile phase of the menstrual cycle?

**A** 2 days **B** 3 days **C** 5 days **D** 7 days

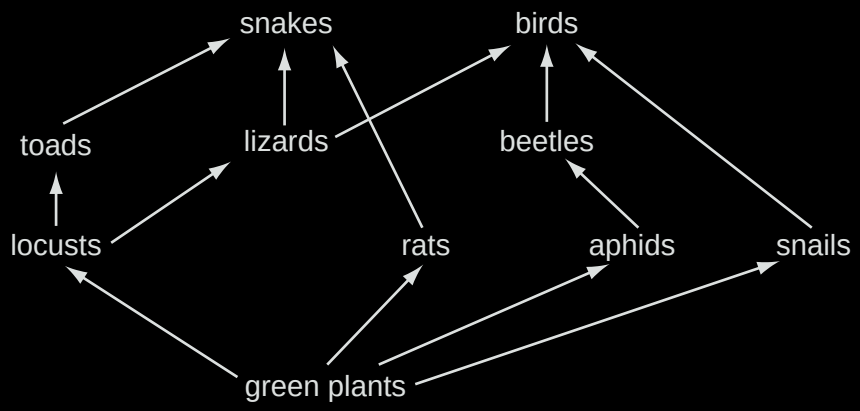
**34** Which **cannot** be an example of excretion?

**A** Carbon dioxide is breathed out from the lungs.

**B** Undigested food leaves the body through the anus.

**C** Urea leaves the body in urine.

**D** Water is removed through the kidneys.




In this food web, the beetles are

**A** carnivores.

**B** decomposers.

**C** herbivores.

**D** producers.

**36** Why is it important to control the amount of nitrate fertiliser used on farm land?

**A** Nitrate causes acid rain which kills trees and fish.

**B** Nitrate decreases the fertility of the soil.

**C** Nitrate may lead to excessive growth of water plants.

**D** Nitrate poisons many kinds of crop plants.

**37** What would be an **undesirable** feature in an insecticide?

**A** It becomes more concentrated at each stage in the food web.

**B** It breaks down within a few months.

**C** It destroys one particular insect only.

**D** It destroys the immature forms of the insect.

**38** Which bacteria convert urea to ammonia?

**A** decay bacteria

**B** denitrifying bacteria

**C** nitrifying bacteria

**D** nitrogen-fixing bacteria